

TP Data Clustering

Exercise 1 :

1)Data analyze using the Jaccard similarity coefficient

We know that :

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} = \frac{|A \cap B|}{|A| + |B| - |A \cap B|}.$$

Since the jaccard similarity is based only on binary values (based on my understanding), we will ignore the sugar column

$$|P1 \cap P2| = 1 \text{ (col 3 only)}$$

$$|P1 \cup P2| = 3$$

$$\text{Donc } J(P1, P2) = 1/3 = 0.3333$$

2)Data analyze using the Jaccard weighted coefficient

We know that :

$$J_W(\mathbf{x}, \mathbf{y}) = \frac{\sum_i \min(x_i, y_i)}{\sum_i \max(x_i, y_i)},$$

$$J_W(P1, P2) = 1+0+0+1 / 2+1+1+1 = 2/5 = 0.4$$

3)Data analyze using the Angular distance

We have :

$$\text{angular distance} = \frac{\cos^{-1}(\text{cosine similarity})}{\pi}$$

If the vector elements are always positive:

$$\text{angular distance} = \frac{2 \cdot \cos^{-1}(\text{cosine similarity})}{\pi}$$

And

$$\text{similarity} = \cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}},$$

Let's first calculate the cosine similarity

$$P1 \times P2 = (1 \times 2) + (1 \times 0) + (0 \times 1) + (1 \times 1) = 3$$

$$||P1|| = \sqrt{1^2 + 1^2 + 0^2 + 1^2} = \sqrt{3}$$

$$||P2|| = \sqrt{2^2 + 0^2 + 1^2 + 1^2} = \sqrt{6}$$

$$\text{Cosine similarity} = 3 / (\sqrt{3} \times \sqrt{6}) = 0.707$$

(since the vector elements are always positive, we will pick the second rule)

$$= (2 \times \cos^{-1}(0.707)) / \pi$$

$$\text{Angular Distance} = 0.5$$

Exercise 2 :

1) Data analyze using the Jaccard similarity coefficient

$$|P1 \cap P2| = 1 \text{ (col 3 only)}$$

$$|P1 \cup P2| = 3$$

$$\text{Donc } J(P1, P2) = 1/3 = 0.3333$$

2) Data analyze using the Jaccard weighted coefficient

$$Jw(P1, P2) = 2+0+0+1 / 3+1+1+1$$

$$Jw(P1, P2) = 3 / 6 = 1/2$$

3) Data analyze using the Angular distance

$$P1 \times P2 = (2 \times 3) + (1 \times 0) + (0 \times 1) + (1 \times 1) = 7$$

$$||P1|| = \sqrt{(2^2 + 1^2 + 0^2 + 1^2)} = \sqrt{6}$$

$$||P2|| = \sqrt{(3^2 + 0^2 + 1^2 + 1^2)} = \sqrt{11}$$

$$\text{Cosine similarity} = 7 / (\sqrt{6} \times \sqrt{11}) = 0.86$$

(since the vector elements are always positive, we will pick the second rule)

$$= (2 \times \cos^{-1}(0.86)) / \pi$$

$$\text{Angular Distance} = 0.33$$

Exercise 3 :

1) K-means with the initial points : A, B and C

We have : {A(2, 2), B(3,3), C(10, 10), D(5, 8), E(8,9), F(5,7), G(1,2)} and know that K = 3, and our initial points are A, B and C

Let's affect M1, M2, M3 for A, B, C which means M1=(2,2), M2 = (3,3) and M3 = (10,10)

D(M1, D)	9
D(M1, E)	13
D(M1, F)	8
D(M1, G)	1

D(M2, D)	7
D(M2, E)	11
D(M2, F)	6
D(M2, G)	3

D(M3, D)	7
D(M3, E)	3
D(M3, F)	8
D(M3, G)	17

(i used an excel sheet to do the calculs)

A	2	2				
B	3	3				
C	10	10				
D	5	8				
E	8	9				
F	5	7				
G	1	2				
D(M1, D)	9		D(M2, D)	7	D(M3, D)	7
D(M1, E)	13		D(M2, E)	11	D(M3, E)	3
D(M1, F)	8		D(M2, F)	6	D(M3, F)	8
D(M1, G)	1		D(M2, G)	3	D(M3, G)	17

Cluster affectations :

D -> C2 or C3

E -> C3

F -> C2

G -> C1

Which means

C1 : A, G

C2 : B, D, F

C3 : C, E

=> M1(1.5, 2)

=> M2(4.33,6)

=> M3(9,9.5)

Let's now repeat the process

D(M1,A)	0.5
D(M1,B)	2.5
D(M1, C)	16.5
D(M1, D)	9.5
D(M1, E)	13.5
D(M1, F)	8.5
D(M1, G)	0.5

D(M2,A)	6.33
D(M2,B)	4.33
D(M2, C)	9.67

D(M2, D)	2.67
D(M2, E)	6.67
D(M2, F)	1.67
D(M2, G)	7.33

D(M3,A)	14.5
D(M3,B)	12.5
D(M3, C)	1.5
D(M3, D)	5.5
D(M3, E)	1.5
D(M3, F)	6.5
D(M3, G)	15.5

A	2	2		M1	1.5	2	
B	3	3		M2	4.33	6	
C	10	10		M3	9	9.5	
D	5	8					
E	8	9					
F	5	7					
G	1	2					
D(M1, D)	9		D(M2, D)	7		D(M3, D)	7
D(M1, E)	13		D(M2, E)	11		D(M3, E)	3
D(M1, F)	8		D(M2, F)	6		D(M3, F)	8
D(M1, G)	1		D(M2, G)	3		D(M3, G)	17
D(M1,A)	0.5		D(M2,A)	6.33		D(M3,A)	14.5
D(M1,B)	2.5		D(M2,B)	4.33		D(M3,B)	12.5
D(M1, C)	16.5		D(M2, C)	9.67		D(M3, C)	1.5
D(M1, D)	9.5		D(M2, D)	2.67		D(M3, D)	5.5
D(M1, E)	13.5		D(M2, E)	6.67		D(M3, E)	1.5
D(M1, F)	8.5		D(M2, F)	1.67		D(M3, F)	6.5
D(M1, G)	0.5		D(M2, G)	7.33		D(M3, G)	15.5

Cluster affectations:

A -> C1

B -> C1

C -> C3

D -> C2

E-> C3

F-> C2

G -> C1

Which means:

C1 : A, B, G

C2 : D, F

C3 : C, E

M1 => (2,2.33)

M2 => (5,7.5)

M3 => (9,9.5)

Let's now repeat the process:

D(M1,A)	0.33333333
D(M1,B)	1.66666667
D(M1, C)	15.6666667
D(M1, D)	8.66666667
D(M1, E)	12.6666667
D(M1, F)	7.66666667
D(M1, G)	1.33333333

D(M2,A)	8.5
D(M2,B)	6.5
D(M2, C)	7.5
D(M2, D)	0.5
D(M2, E)	4.5
D(M2, F)	0.5
D(M2, G)	9.5

D(M3,A)	14.5
D(M3,B)	12.5
D(M3, C)	1.5
D(M3, D)	5.5
D(M3, E)	1.5
D(M3, F)	6.5
D(M3, G)	15.5

C1 : A, B, G

C2 : D, F

C3 : C, E

Clusters remain the same so the final results are :

M1	2	2.33333333
M2	5	7.5
M3	9	9.5

2) Agglomerative clustering with single linkage:
Here are the steps of clustering following the minimum

	A	B	C	D	E	F	G
A	0						
B	2	0					
C	16	14	0				
D	9	7	7	0			
E	13	11	3	4	0		
F	8	6	8	1	5	0	
G	1	3	17	10	14	9	0

	AG	B	C	D	E	F
AG	0					
B	2	0				
C	16	14	0			
D	9	7	7	0		
E	13	11	3	4	0	
F	8	6	8	1	5	0

	AG	B	C	DF	E
AG	0				
B	2	0			
C	16	14	0		
DF	8	6	7	0	
E	13	11	3	4	0

	AGB	C	DF	E
AGB	0			
C	14	0		
DF	6	7	0	
E	11	3	4	0

	AGB	CE	DF
AGB	0		
CE	11	0	
DF	6	7	0

3) Agglomerative clustering with complete linkage

	A	B	C	D	E	F	G
A	0						
B	2	0					
C	16	14	0				
D	9	7	7	0			
E	13	11	3	4	0		
F	8	6	8	1	5	0	
G	1	3	17	10	14	9	0

	AG	B	C	D	E	F
AG	0					
B	3	0				
C	17	14	0			
D	10	7	7	0		
E	14	11	3	4	0	
F	9	6	8	1	5	0

	AG	B	C	DF	E
AG	0				
B	3	0			
C	17	14	0		
DF	10	7	8	0	
E	14	11	3	5	0

	AG	B	CE	DF
AG	0			
B	3	0		
CE	17	14	0	
DF	10	7	8	0

	ABG	CE	DF
AG	0		
CE	17	0	
DF	10	8	0

4) Agglomerative clustering with average linkage.

A	B	C	D	E	F	G
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A	0	2	16	9	13	8	1
B	2	0	14	7	11	6	3
C	16	14	0	7	3	8	17
D	9	7	7	0	4	1	10
E	13	11	3	4	0	5	14
F	8	6	8	1	5	0	9
G	1	3	17	10	14	9	0
DistAvg	8.16666667	7.16666667	10.83333333	6.33333333	8.33333333	6.16666667	9

	A	B	D	E	F	G
A	0	2	9	13	8	1
B	2	0	7	11	6	3
D	9	7	0	4	1	10
E	13	11	4	0	5	14
F	8	6	1	5	0	9
G	1	3	10	14	9	0
DistAvg	6.6	5.8	6.2	9.4	5.8	7.4
Dist(C)	16	14	7	3	8	17

	A	B	D	F	G
A	0	2	9	8	1
B	2	0	7	6	3
D	9	7	0	1	10
F	8	6	1	0	9
G	1	3	10	9	0
DistAvg	5	4.5	6.75	6	5.75
Dist(CE)	13	11	4	5	14

	A	B	F	G
A	0	2	8	1
B	2	0	6	3
F	8	6	0	9
G	1	3	9	0
DistAvg	3.66666667	3.66666667	7.66666667	4.33333333
Dist(CED)	9	7	1	10

	A	B	G
A	0	2	1
B	2	0	3

G	1	3	0
DistAvg	1.5	2.5	2
Dist(CEDF)	8	6	9

	A	G
A	0	1
G	1	0
DistAvg	1	1
Dist(B)	2	3
Dist(CEDF)	8	9

So the 3 clusters are : AG, B, CEDF